

1 **Half-Plane Shielding: Provable Directional COLREGs Compliance for Continuous-Action Safe**
2 **Reinforcement Learning**

3 **Weiyu Tang**

4 Graduate Research Assistant
5 Shanghai Key Laboratory for Digital Maintenance of Buildings and Infrastructure,
6 School of Naval Architecture, Ocean and Civil Engineering
7 Shanghai Jiao Tong University, Shanghai, 200240, China
8 Email: weiyutang@sjtu.edu.cn

9 **Jie Xue, Ph.D.***

10 Assistant Research Professor
11 State Key Laboratory of Ocean Engineering
12 School of Naval Architecture, Ocean and Civil Engineering
13 Shanghai Jiao Tong University, Shanghai, 200240, China
14 Email: j.xue@sjtu.edu.cn

15 **Peijie Yang**

16 Shanghai Key Laboratory for Digital Maintenance of Buildings and Infrastructure,
17 School of Naval Architecture, Ocean and Civil Engineering
18 Shanghai Jiao Tong University, Shanghai, 200240, China
19 Email: yangpeijie21@sjtu.edu.cn

20 **Qianbing Li**

21 Shanghai Key Laboratory for Digital Maintenance of Buildings and Infrastructure,
22 School of Naval Architecture, Ocean and Civil Engineering
23 Shanghai Jiao Tong University, Shanghai, 200240, China
24 Email: qianbing_li@sjtu.edu.cn

25 Word Count: [TBD] words + [TBD] table(s) $\times$ 250 = [TBD] words

26 Total Number of Pages: [TBD]

27 Submitted [Submission Date]

28 *Corresponding Author

1 ABSTRACT

2 **Objectives.** Autonomous surface vessels must obey the International Regulations for Preventing Collisions
3 at Sea (COLREGs). The give-way rules are directional. A track that avoids a collision but passes on
4 the wrong side is still a violation. This paper builds a per-step guarantee of directional compliance for
5 continuous control commands.

6 **Methods.** The give-way clause is written as a half-plane constraint on the plane of control com-
7 mands. It is intersected with the actuator limits and a one-step collision-free condition, which gives a convex
8 safe control set. A small quadratic program projects the command of the policy onto the nearest point of
9 that set. The projection sits inside the environment transition, so the policy learns under its own corrected
10 commands. The data are 2,000 simulated two-vessel encounters, split into 1,400 training and 600 test cases.
11 Nine configurations form a single-variable ladder, each trained with eight fixed seeds.

12 **Findings.** On every give-way step where the projection is feasible, directional compliance holds by
13 construction. It does not depend on the training outcome or the seed. Steps that switch to the fallback branch
14 are not covered. On the test set, the violation count per episode is [TBD] and the collision rate is [TBD].
15 The ablation gains are significant under a paired signed-rank test. The arrival rate is not claimed as a benefit.

16 **Novelty.** To the best of the authors' knowledge, this is the first projection-based safety mechanism
17 that enforces directional COLREGs compliance on continuous control commands. The paper also states
18 what is proved and what is not.

19 **Practical Applications.** The compliance check is independent of training. The mechanism can sit
20 above an existing controller as a compliance layer for functional verification and liability assessment. The
21 online cost is one small quadratic program per step, so it runs in real time without special hardware.

22 **Keywords:** COLREGs compliance, safe reinforcement learning, projection-based safety shield, continuous
23 action space, quadratic programming, unmanned surface vehicle

1 METHODOLOGY

2 The Give-Way Clause as a Half-Plane

3 The rule state machine sorts the current encounter into six situations,

$$\rho \in \{\rho_0 \text{ conflict-free}, \rho_1 \text{ stand-on}, \rho_2 \text{ head-on}, \rho_3 \text{ crossing}, \rho_4 \text{ overtaking}, \rho_5 \text{ emergency}\},$$

4 and the decision is a conjunction of three groups of predicates.

5 The first group is the bearing sector, taken along the sidelight boundaries of the convention. The
 6 second group is the heading relation, which separates nearly anti-parallel from nearly parallel headings. The
 7 third group is collision possibility. It has two parts. The speed set of the own vessel must intersect the
 8 collision cone of the target vessel. The relative speed must also close the current center distance in time.
 9 The forward sector has a half-angle of 5° and the side sectors end at 112.5° . The overtaking band has a
 10 half-angle of 67.5° . The occupancy of the target vessel is inflated by three times its length, and the check
 11 horizon is 420 s.

12 Four situations carry an obligation,

$$\begin{aligned} \rho_2 \text{ head-on : } & \text{CP} \wedge \text{Fr}(s_e, s_m) \wedge \text{Ap}, \\ \rho_3 \text{ crossing : } & \text{CP} \wedge \text{Ri}(s_e, s_m) \wedge \text{TI}, \\ \rho_4 \text{ overtaking : } & \text{CP} \wedge \text{Be}(s_m, s_e) \wedge (v_e > v_m) \wedge \text{Sd}, \\ \rho_1 \text{ stand-on : } & [\text{CP} \wedge \text{Le}(s_e, s_m) \wedge \text{Tr}] \vee \rho_4(s_m, s_e), \end{aligned} \quad (1)$$

13 and every other case falls into ρ_0 . Overtaking and being overtaken both swap the two vessel arguments,
 14 because overtaking places the own vessel in the aft sector of the target vessel. The four predicates can hold
 15 at the same time in geometry. They are therefore applied in the order written above.

16 Transitions are triggered by a persistence condition. Entry from the conflict-free situation into a
 17 give-way situation has two parts. A give-way predicate must not hold now. It must also hold at every
 18 decision step of a 60 s reaction window under constant-velocity extrapolation. The first part separates
 19 “about to hold” from “already holds”. It also creates one asymmetry, which Section treats. The emergency
 20 situation is entered by a stronger set-based prediction with a horizon of 180 s.

21 **Compliant control set.** Each situation maps to an explicit constraint set on the command,

$$U_{\text{colregs}}(\rho) = \begin{cases} U_{\text{box}}, & \rho = \rho_0, \\ \{|a| \leq \epsilon_a\} \cap \{|\omega| \leq \epsilon_\omega\}, & \rho = \rho_1, \\ \{\omega \leq -\omega_{\text{turn}}\}, & \rho \in \{\rho_2, \rho_3\}, \\ \{\omega \leq -\omega_{\text{turn}}\} \text{ or } \{\omega \geq +\omega_{\text{turn}}\}, & \rho = \rho_4, \\ U_{\text{box}}, & \rho = \rho_5. \end{cases} \quad (2)$$

22 Head-on and crossing require a turn to starboard. Overtaking allows a turn to either side. Stand-on requires
 23 the course and the speed to be held. Each of these sets is a half-plane or an interval on the plane of control
 24 commands. That shape is what the rest of this paper builds on. A set of this shape can be entered by
 25 projection, so the commands do not have to be enumerated.

26 **Quantifying a readily apparent action.** The convention asks a give-way action to be large enough to be
 27 seen in time. We quantify this as a course change of at least a given threshold within the maneuver window.
 28 With a window of 40 s and a threshold of 20° ,

$$\omega_{\text{turn}} = \frac{\Delta_{\text{lt}}}{T_M} = \frac{20^\circ}{40 \text{ s}} \approx 0.008727 \text{ rad/s}. \quad (3)$$

This is the infimum of the compliant turn rate. Any turn at least this large complies, and the bound cannot be lowered.

The structure and the thresholds of the predicates are taken from the temporal-logic formalization of the maritime rules (Krasowski and Althoff 2024). This paper adds three things. The inflation radius and the half-width of the speed set are stated explicitly. The entry into a give-way situation is made symmetric (Section). The predicate set is then connected to a constraint set on continuous commands.

Safe Control Set and Projection

One-Step Collision-Free Constraint

At each decision step the applied command must keep the two hulls apart at the end of the next step. Written directly, that condition is not convex. We therefore use a conservative convex inner approximation. The construction has three parts.

First, the occupancy of the target vessel after one step is over-approximated by a disk. Its center follows a constant-velocity extrapolation. Its radius already contains a safety margin. The own vessel is over-approximated by its circumscribed circle. A sufficient condition for the two hulls to stay apart is then simple. The distance between the two centers must be at least the sum of the two radii, written d_{safe} .

Second, let n be the unit vector from the target vessel to the nominal end point of the own vessel, and let δ be that distance. The nominal end point follows from the policy command after clipping to the action box.

Third, the end point is expanded to first order in the command, $p_e(u) \approx p_{\text{nom}} + J(u - u_{\text{nom}})$. The end point must lie on the safe side of the separating hyperplane. This gives one linear inequality in the command:

$$\underbrace{-n^\top J u}_{g^\top} \leq \underbrace{-(d_{\text{safe}} - \delta + n^\top J u_{\text{nom}})}_h, \quad U_{\text{cf}}(s) = \{u : g^\top u \leq h\}. \quad (4)$$

A first-order expansion is neither an inner nor an outer approximation. The sign of the remainder changes with the curvature. Equation (4) alone therefore does not establish conservatism. The margin inside the safety distance is what does. With that margin the constraint acts at a center distance of about 590–840 m. The median over this scenario library is about 714 m. Hull contact needs a much smaller distance. The margin thus absorbs the linearization residual. The cost is a tighter constraint. Some commands that are in fact safe are rejected.

The linearization in Equation (4) holds only locally. In the far field a sufficient condition is available that does not rely on it.

Property 1 (One-step collision freedom in the far field). *Let d be the center distance between the own vessel and a single target vessel. If*

$$d \geq d_{\text{safe}} + v_{\text{obs,max}} \Delta t + \rho_{\text{ego}}, \quad (5)$$

then the two conservative circumscribed occupancies do not intersect after one decision step. Here d_{safe} is the sum of the two circumscribed radii, and ρ_{ego} is the largest displacement of the own vessel in one step.

Assumptions. A single target vessel. The speed of the target vessel is at most 9.5 m/s. That bound is not implied by the dynamics and is stated separately; the largest value measured over the 2,000 scenarios of the library is 7.10 m/s. The target vessel holds its speed within the step.

Proof sketch. By the triangle inequality, the center distance after the step is at least d minus the one-step displacement bound of each vessel. It is therefore at least d_{safe} .

Scope. The triangle inequality does not depend on direction, so a head-on encounter is the worst case. With the constants of this library the threshold is about 764 m. The statement covers one step. It is not forward invariance, and it does not cover the near field.

1 *Safe Control Set and the Projection Operator*

2 The safe control set is the intersection of three sets,

$$U_{\text{safe}}(s, \rho) = U_{\text{box}} \cap U_{\text{colregs}}(\rho) \cap U_{\text{cf}}(s), \quad (6)$$

3 and it is a convex polygon in $\mathbb{R}^2$. The applied command is the Euclidean projection of the policy command
 4 onto that set,

$$u_{\text{safe}} = \Pi_{U_{\text{safe}}}(u_{\text{policy}}) = \arg \min_{u \in U_{\text{safe}}} \|u - u_{\text{policy}}\|_2^2. \quad (7)$$

5 The feasible set is an intersection of half-planes. It can be written as $U_{\text{safe}} = \{u : Au \leq b\}$. The
 6 first four rows are the action box. The fifth row is the compliant-direction half-plane. The sixth row is
 7 the collision-free constraint of Equation (4). This is a quadratic program in two variables with at most six
 8 constraints. It is small enough to solve online at every decision step.

9 This construction gives the central guarantee of the paper.

10 **Property 2** (Directional compliance on every give-way step). *Consider any decision step that the rule state*
 11 *machine assigns to a give-way situation and on which the projection is feasible. The applied command u_{safe}*
 12 *then lies in the compliant-direction half-plane.*

13 **Assumptions.** A single target vessel, and a correct decision by the state machine. The second assumption
 14 inherits one conservative bias of the collision-possibility predicate. That predicate uses the center distance,
 15 and it can miss a trailing encounter in which the two speeds are close.

16 **Proof sketch.** The compliant half-plane enters the quadratic program as a hard constraint. Any feasible
 17 point satisfies it, and the projection returns a feasible point. The statement holds by construction. It does not
 18 depend on the training outcome or on the random seed.

19 **Scope.** Emergency situations are not covered, because their branch bypasses the compliance constraint by
 20 design. Steps on which the safe control set is empty and the fallback is used are not covered either.

21 **REFERENCES**

- 22 Alshiekh, Mohammed, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk
 23 Topcu. 2018. “Safe Reinforcement Learning via Shielding.” In *Proceedings of the AAAI Conference*
 24 *on Artificial Intelligence*.
 25 Ames, Aaron D., Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo
 26 Tabuada. 2019. “Control Barrier Functions: Theory and Applications.” In *European Control Con-*
 27 *ference (ECC)*, 3420–3431.
 28 Benjamin, Michael R., John J. Leonard, Joseph A. Curcio, and Paul M. Newman. 2006. “A Method for
 29 Protocol-Based Collision Avoidance Between Autonomous Marine Surface Craft.” *Journal of Field*
 30 *Robotics* 23 (5): 333–346.
 31 Cheng, Yin, and Weidong Zhang. 2018. “Concise Deep Reinforcement Learning Obstacle Avoidance for
 32 Underactuated Unmanned Marine Vessels.” *Neurocomputing* 272: 63–73.
 33 Chou, Po-Wei, Daniel Maturana, and Sebastian Scherer. 2017. “Improving Stochastic Policy Gradients in
 34 Continuous Control with Deep Reinforcement Learning Using the Beta Distribution.” In *Proceed-*
 35 *ings of the International Conference on Machine Learning (ICML)*.
 36 Chun, D.-H., M.-I. Roh, H.-W. Lee, J. Ha, and D. Yu. 2021. “Deep Reinforcement Learning-Based Collision
 37 Avoidance for an Autonomous Ship.” *Ocean Engineering* 234: 109216.
 38 Dalal, Gal, Krishnamurthy Dvijotham, Matej Vecerik, Todd Hester, Cosmin Paduraru, and Yuval Tassa.
 39 2018. “Safe Exploration in Continuous Action Spaces.” arXiv:1801.08757.
 40 De Lellis, Francesco, Marco Coraggio, Giovanni Russo, Mirco Musolesi, and Mario di Bernardo. 2024.
 41 “Guaranteeing Control Requirements via Reward Shaping in Reinforcement Learning.” *IEEE*
 42 *Transactions on Control Systems Technology*. arXiv:2311.10026.

- 1 Eriksen, Bjørn-Olav H., Morten Breivik, Erik F. Wilthil, Andreas L. Flåten, and Edmund F. Brekke. 2019.
- 2 “The Branching-Course Model Predictive Control Algorithm for Maritime Collision Avoidance.”
- 3 *Journal of Field Robotics* 36.
- 4 Fossen, Thor I. 2011. *Handbook of Marine Craft Hydrodynamics and Motion Control*. Chichester: John
- 5 Wiley & Sons.
- 6 García, Javier, and Fernando Fernández. 2015. “A Comprehensive Survey on Safe Reinforcement Learn-
- 7 ing.” *Journal of Machine Learning Research* 16: 1437–1480.
- 8 Grover, Jaskaran, Changliu Liu, and Katia Sycara. 2023. “The Before, During, and After of Multi-Robot
- 9 Deadlock.” *International Journal of Robotics Research*. arXiv:2206.01781.
- 10 Heiberg, Amalie, Thomas Nakken Larsen, Eivind Meyer, Adil Rasheed, Omer San, and Damiano Varagnolo.
- 11 2022. “Risk-Based Implementation of COLREGs for Autonomous Surface Vehicles Using Deep
- 12 Reinforcement Learning.” *Neural Networks* 152: 17–33. arXiv:2112.00115.
- 13 Huang, Yamin, Linying Chen, and P. H. A. J. van Gelder. 2019. “Generalized Velocity Obstacle Algorithm
- 14 for Preventing Ship Collisions at Sea.” *Ocean Engineering* 173: 142–156.
- 15 IMO (International Maritime Organization). 1972. *Convention on the International Regulations for Pre-*
- 16 *venting Collisions at Sea (COLREGs)*. London: IMO.
- 17 Johansen, Tor A., Tristan Perez, and Andrea Cristofaro. 2016. “Ship Collision Avoidance and COLREGS
- 18 Compliance Using Simulation-Based Control Behavior Selection with Predictive Hazard Assess-
- 19 ment.” *IEEE Transactions on Intelligent Transportation Systems* 17 (12): 3407–3422.
- 20 Kim, Kyungmin, Davide Corsi, Andoni Rodríguez, JB Lanier, Benjamí Parellada, Pierre Baldi, César
- 21 Sánchez, and Roy Fox. 2024. “Realizable Continuous-Space Shields for Safe Reinforcement Learn-
- 22 ing.” arXiv:2410.02038.
- 23 Kochdumper, Niklas, Hanna Krasowski, Xiao Wang, Stanley Bak, and Matthias Althoff. 2023. “Provably
- 24 Safe Reinforcement Learning via Action Projection Using Reachability Analysis and Polynomial
- 25 Zonotopes.” *IEEE Open Journal of Control Systems* 2.
- 26 Krasowski, Hanna, and Matthias Althoff. 2022. “CommonOcean: Composable Benchmarks for Mo-
- 27 tion Planning on Oceans.” In *Proceedings of the IEEE International Conference on Intelligent*
- 28 *Transportation Systems (ITSC)*, 1676–1682. DOI: 10.1109/ITSC55140.2022.9921925. [https:](https://commonocean.cps.cit.tum.de/)
- 29 [//commonocean.cps.cit.tum.de/](https://commonocean.cps.cit.tum.de/).
- 30 Krasowski, Hanna, and Matthias Althoff. 2024. “Provable Traffic Rule Compliance in Safe Reinforce-
- 31 ment Learning on the Open Sea.” *IEEE Transactions on Intelligent Vehicles* 9 (12): 7617–7634.
- 32 arXiv:2402.08502.
- 33 Kufoalor, D. K. M., Tor A. Johansen, Edmund F. Brekke, A. Hepsø, and K. Trnka. 2020. “Autonomous
- 34 Maritime Collision Avoidance: Field Verification of Autonomous Surface Vehicle Behavior in Chal-
- 35 lenging Scenarios.” *Journal of Field Robotics* 37.
- 36 Kuwata, Yoshiaki, Michael T. Wolf, Dimitri Zarghitsky, and Terrance L. Huntsberger. 2014. “Safe Maritime
- 37 Autonomous Navigation with COLREGS, Using Velocity Obstacles.” *IEEE Journal of Oceanic*
- 38 *Engineering* 39 (1): 110–119.
- 39 Lee, Changyu, Jinwook Park, and Jinwhan Kim. 2025. “Efficient COLREGs-Compliant Collision Avoid-
- 40 ance Using Turning Circle-Based Control Barrier Function.” arXiv:2504.19247.
- 41 Markgraf, Hannah, Shambhuraj Sawant, Hanna Krasowski, Lukas Schäfer, Sébastien Gros, and Matthias
- 42 Althoff. 2026. “Safe Reinforcement Learning Using Action Projection: Safeguard the Policy or the
- 43 Environment?” *Transactions on Machine Learning Research*. arXiv:2509.12833.
- 44 Meyer, Eivind, Amalie Heiberg, Adil Rasheed, and Omer San. 2020. “COLREG-Compliant Collision
- 45 Avoidance for Unmanned Surface Vehicle Using Deep Reinforcement Learning.” *IEEE Access* 8.
- 46 arXiv:2006.09540.
- 47 Müller, Henrik, and Daniel Kudenko. 2025. “Improving the Effectiveness of Potential-Based Reward Shap-
- 48 ing in Reinforcement Learning.” In *Proceedings of the International Conference on Autonomous*

- 1 *Agents and Multiagent Systems (AAMAS)*. arXiv:2502.01307.
- 2 Müller, Marlon, Florian Finkeldei, Hanna Krasowski, Murat Arcak, and Matthias Althoff. 2026.
- 3 “Falsification-Driven Reinforcement Learning for Maritime Motion Planning.” *Ocean Engineer-*
- 4 *ing* 361: 125579. arXiv:2510.06970.
- 5 Ng, Andrew Y., Daishi Harada, and Stuart Russell. 1999. “Policy Invariance Under Reward Transforma-
- 6 tions: Theory and Application to Reward Shaping.” In *Proceedings of the International Conference*
- 7 *on Machine Learning (ICML)*, 278–287.
- 8 Oh, Donggeon David, Duy P. Nguyen, Haimin Hu, and Jaime F. Fisac. 2025. “Provably Optimal Reinforce-
- 9 ment Learning under Safety Filtering.” arXiv:2510.18082.
- 10 Patil, Mayur S., Nataraj Sudharsan, Veneela Ammula, Jude Tomdio, Jin Wang, Michael Kei, Sivaku-
- 11 mar Rathinam, and Prabhakar R. Pagilla. 2026. “COLREGs Compliant Collision Avoidance and
- 12 Grounding Prevention for Autonomous Marine Navigation.” arXiv:2603.02484.
- 13 Pizarro Bejarano, Federico, Lukas Brunke, and Angela P. Schoellig. 2025. “Safety Filtering While Train-
- 14 ing: Improving the Performance and Sample Efficiency of Reinforcement Learning Agents.” *IEEE*
- 15 *Robotics and Automation Letters*. arXiv:2410.11671.
- 16 Raffin, Antonin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann.
- 17 2021. “Stable-Baselines3: Reliable Reinforcement Learning Implementations.” *Journal of Machine*
- 18 *Learning Research* 22 (268): 1–8.
- 19 Sarhadi, Pouria, Wasif Naeem, and Nikolaos Athanasopoulos. 2022. “A Survey of Recent Machine
- 20 Learning Solutions for Ship Collision Avoidance and Mission Planning.” *IFAC-PapersOnLine*.
- 21 arXiv:2207.02767.
- 22 Schulman, John, Philipp Moritz, Sergey Levine, Michael I. Jordan, and Pieter Abbeel. 2016. “High-
- 23 Dimensional Continuous Control Using Generalized Advantage Estimation.” In *International Con-*
- 24 *ference on Learning Representations (ICLR)*.
- 25 Schulman, John, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. “Proximal Policy
- 26 Optimization Algorithms.” arXiv:1707.06347.
- 27 Shen, Haiqing, Hirotada Hashimoto, Akihiko Matsuda, Yuuki Taniguchi, Daisuke Terada, and Chen Guo.
- 28 2019. “Automatic Collision Avoidance of Multiple Ships Based on Deep Q-Learning.” *Applied*
- 29 *Ocean Research* 86: 268–288.
- 30 Trott, Alexander, Stephan Zheng, Caiming Xiong, and Richard Socher. 2019. “Keeping Your Distance:
- 31 Solving Sparse Reward Tasks Using Self-Balancing Shaped Rewards.” In *Advances in Neural In-*
- 32 *formation Processing Systems (NeurIPS)*.
- 33 Vaaler, Aksel, Svein Jostein Husa, Daniel Menges, Thomas Nakken Larsen, and Adil Rasheed. 2024.
- 34 “Modular Control Architecture for Safe Marine Navigation: Reinforcement Learning and Predictive
- 35 Safety Filters.” arXiv:2312.01855.
- 36 Wabersich, Kim P., and Melanie N. Zeilinger. 2021. “A Predictive Safety Filter for Learning-Based Control
- 37 of Constrained Nonlinear Dynamical Systems.” *Automatica* 129: 109597.
- 38 Zhao, Luman, and Myung-II Roh. 2019. “COLREGs-Compliant Multiship Collision Avoidance Based on
- 39 Deep Reinforcement Learning.” *Ocean Engineering* 191: 106436.